Dr. Babasaheb Ambedkar Technological University, Lonere - Maharashtra

NAGESH KARAJAGI *ORCHID* COLLEGE OF ENGINEERING & TECHNOLOGY, SOLAPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Final year Project Work Diary

Academic Year: 2022-2023

Project Title: Design & development of an interactive website and chatbot for students preparing for interviews and performance evaluation.

Project Group No: A- 2

Project Group Members:

Sr No.	Name	Roll No.	Exam Seat No.
1	Ms. YASHASHWINI AWATE	05	1962231242051
2	Ms. GUNAJA LIGADE	06	1962231242143
3	Ms. SAYALI KURRA	07	1962231242145
4	Mr. ABHAY PATHAK	08	1962231242058

Project Guide: DR. V.V. BAG

Project Schedule

Semester-1

Month	Week	Project Activity
September	Week 4	Formation of the project group
October	Week 1	Finalization of the project idea
	Week 2	Finalization of the project title
	Week 3	Study of existing systems.
	Week 4	Literature Survey, Literature Review.
November	Week 1	Project Abstract.
	Week 2	Model of Implementation.
	Week 3	Proposed Work.
	Week 4	Requirement Analysis.
December	Week 1	System Architecture.
	Week 2	Activity Diagram, Sequence Diagram, Class Diagram.
	Week 3	Report writing, Project Diary writing.
	Week 4	Report Submission

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Final year Project Work Diary

Academic Year: 2022-2023

Project Title: Development of Machine learning Model using Gradio

Project Group No: A- 2

Project Group Members:

Sr No.	Name	Roll No.	Exam Seat No.
1	Ms. YASHASHWINI AWATE	05	1962231242051

Project Guide: DR. V.V. BAG

Semester – II

Project Schedule:

Month	Week	Activity
March	Week 1	50% remaining form design completion
	Week 2	Review on forms and reports
	Week 3	Form implementation
	Week 4	Gather feedback
April	Week 1	**Testing of application**
	Week 2	Bug fixing after testing
	Week 3	Deployment of application
		User Manual Implementation(help files)
	Week 4	Perform error analysis
May	Week 1	Client approval after deployment, letter submission
		Report writing
	Week 2	Necessary refinements
	Week 3	Deployment
	Week 4	Application's performance and user feedback

Project Progress Sheet

Semester – I

Activity week: September Week 4 From 23/09/2022 to 30/09/2022

Activity according to schedule:

1. Formation of the project group

Group is formed with four members in a team.

Date: 25/09/2022 Guide

Activity week: October Week 1 From 01/10/2022 to 07/10/2022

Activity according to schedule:

1. Finalization of the project idea Work

done:

Choosing the right final year project topic holds paramount importance, so according to the demand and requirement in society and the changes it can bring, the project idea was finalized based on the study/research of society.

Date: 05/10/2022 Guide

Activity week: October Week 2 From 07/10/2022 to 14/10/2022

Activity according to schedule:

1. Finalization of project title

According to the research done on the requirements of society and considering the project background, the project title is finalized as "DESIGN & DEVELOPMENT OF INTERACTIVE WEBSITE AND CHATBOT FOR STUDENTS PREPARING FOR INTERVIEWS, ALONG WITH PERFORMANCE EVALUATION."

Date: 12/10/2022 Guide

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Activity week: October Week 3 From 14/10/2022 to 21/10/2022

Activity according to schedule:

1. Study of existing systems

Work done:

Studied/Researched the existing system and analyzed the gap of difference in the form of many different required services.

Date: 18/10/2022 Guide

Activity week: October Week 4 From 21/10/2022 to 28/10/2022

Activity according to schedule:

1. Literature survey and . Literature review.

Work done:

Through much research and understanding of the requirements of the project as well as the system, the literature survey is completed. The literature review is formed according to the literature survey and requirements.

Date: 25/10/2022 Guide

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Activity week: November Week 1 From 28/10/2022 to 04/11/2022

Activity according to schedule:

1. Project Abstract

Work done:

The abstract is formed by understanding the system's requirements, highlighting the significant points of research, and explaining why this project is essential, what the purpose is, actual workflow, learning, and conclusion.

Date: 1/11/2022 Guide

Activity week: November Week 2 From 04/11/2022 to 11/11/2022

Activity according to schedule:

1. Model of Implementation.

Work done:

According to the literature, research, abstract, and understanding of the workflow requirement, the Model of Implementation of the project is completed.

Date: 7/11/2022 Guide

Activity week: November Week 3 From 11/11/2022 to 18/11/2022

Activity according to schedule:

1. Proposed work

Work done:

According to the literature, research, abstract, and understanding of the workflow requirement, the analysis of the project is completed.

Date: 15/11/2022 Guide

Activity week: November Week 4 From 18/11/2022 to 25/11/2022

Activity according to schedule:

1. Requirement Analysis

Work done

Considering the project's workflow, requirements, and modules, system architecture diagram the requirement Analysis of hardware and software required are completed.

Date: 21/11/2022 Guide

Activity week: December Week 1 From 25/11/2022 to 02/12/2022

Activity according to schedule:

1. System Architecture Work

done:

Considering the project's workflow and step-by-step implementation and workflow, requirements, and modules, system architecture is formed.

Date: 27/11/2022 Guide

Activity week: December Week 2 From 02/12/2022 to 09/12/2022

Activity according to schedule:

1. Activity Diagram, Sequence Diagram, Class Diagram.

Work done

All diagrams are completed by understanding/studying system architecture and algorithms and considering the required modules.

Date: 5/12/2022 Guide

Activity week: December Week 3 From 09/12/2022 to 16/12/2022

Activity according to schedule:

1. Report writing
2. Work Diary

Work done:

Report writing and work diary formation are completed by considering the existing system, abstract, literature review, etc., and based on work done.

Date: 13/12/2022 Guide

Activity week: December Week 4 From 16/12/2022 to 23/12/2022

Activity according to schedule:

Report Submission Work

done:

The final report is submitted by following all instructions with the completion of the report.

Date: 20/12/2022 Guide

Project Coordinator **HOD**

Project Progress Sheet

Semester – II

Activity week: March Week 1

Work done:

1. Research and familiarize yourself with the problem of flower image recognition using machine learning.
2. Gather and study relevant datasets of flower images.
3. Set up the development environment and install the necessary libraries and frameworks, such as TensorFlow or PyTorch.
4. Build a basic image classification pipeline using a pre-trained model for initial experimentation.

Date: 6-3-2023 Guide

Activity week: March Week 2

Review on forms and reports

Work done:

1. Preprocess the flower image dataset by resizing, normalizing, and augmenting the images as required.
2. Explore different pre-trained models suitable for image classification tasks and choose the most appropriate one.
3. Fine-tune the selected pre-trained model on the flower dataset.
4. Evaluate the model's performance using appropriate metrics and make necessary adjustments.

Date: 13-3-2023 Guide

Activity week: March Week 3

Form implementation

Work done:

1. Implement a basic user interface using Gradio to create an interactive environment for users to upload and classify flower images.
2. Integrate the trained model with the Gradio interface.
3. Test the interface and ensure it functions correctly by uploading sample flower images and obtaining predictions

Date: 20-3-2023 Guide

Activity week: March Week 4

Gather feedback

Work done:

1. Gather feedback on the initial Gradio interface and make necessary improvements based on user suggestions.
2. Experiment with different image preprocessing techniques to enhance the model's performance.
3. Optimize hyperparameters of the model to improve its accuracy and efficiency.
4. Perform cross-validation to validate the model's performance and generalizability.

Date: 27-3-2023 Guide

Project Coordinator HOD

Activity week: April Week 1

Testing of application

1. Work done: Explore techniques such as transfer learning to improve the model's performance further.
2. Experiment with different architectures and ensembling methods to enhance the model's accuracy.
3. Explore techniques to handle class imbalances in the dataset if present.
4. Evaluate and compare the performance of different models and architectures.

Date: 3-4-2023 Guide

Activity week: April Week 2

1. Bug fixing after testing

Work done:

1. Refine the user interface to enhance its usability and aesthetics.
2. Implement functionality for displaying the top predicted classes along with probabilities.
3. Add error handling and edge case scenarios to improve the robustness of the application.
4. Conduct extensive testing to ensure the system is stable and reliable.

Date: 10-4-2023 Guide

Activity week: April Week 3

Deployment of application

User Manual Implementation(help files)

1. Work done: Refine the user interface to enhance its usability and aesthetics.
2. Implement functionality for displaying the top predicted classes along with probabilities.
3. Add error handling and edge case scenarios to improve the robustness of the application.
4. Conduct extensive testing to ensure the system is stable and reliable.

Date: 17-4-2024 Guide

Activity week: April Week 4

Perform error analysis

Work done:

1. Perform error analysis to identify common misclassifications or issues.
2. Fine-tune the model based on the error analysis findings.
3. Investigate potential techniques to improve the interpretability of the model's predictions.
4. Optimize the user interface for better responsiveness and user experience.

Date: 24-4-2023 Guide

Project Coordinator HOD

Activity week: May Week 1

Activity according to schedule:

1. approval after deployment, Sponsorship letter submission.
2. Report writing

Work done:

1. Explore techniques for model compression and quantization to reduce the model's size and improve its deployment efficiency.
2. Implement caching mechanisms to enhance the application's performance and reduce inference time.
3. Conduct performance testing to identify any bottlenecks and optimize the system accordingly.
4. Document the project, including the model architecture, implementation details, and user instructions.

Date: 1-5-2023 Guide

Activity week: May Week 2

necessary refinements

Work done:

1. Perform a final round of testing and debugging to ensure the project is free from major issues.
2. Collect additional user feedback on the application's usability and functionality.
3. Make necessary refinements and improvements based on user feedback.
4. Prepare the project for deployment and consider options for hosting the application.

Date: 8-5-2023 Guide

Activity week: May Week 3

Deployment

Work done:

1. Deploy the flower image recognition application to a cloud platform or server.
2. Set up necessary infrastructure, including hosting, scaling, and monitoring.
3. Conduct load testing to ensure the application can handle multiple concurrent users efficiently.
4. Finalize any remaining user interface improvements and polish the application.

Date: 16-5-2023 Guide

Activity week: May Week 4

Application's performance and user feedback.

Work done:

1. Conduct a comprehensive evaluation of the deployed application's performance and user feedback.
2. Collect usage analytics and analyze them to gain insights for further improvements.
3. Address any reported bugs or issues in the deployed system promptly.
4. Prepare for a potential public release or sharing the application with a wider audience.

Date: 21-5-2023 Guide

Project Coordinator HOD